

One Ocean, Many Tasks: A Data-Grounded Simulation Suite and Benchmark for Marine Robotics

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Abstract

Simulation is essential for marine robotics, yet existing platforms often trade off between visually realistic worlds, high-throughput experimentation, and ocean-condition realism grounded in real data. We present OneOcean, a data-grounded simulation suite and benchmark built around a unified spatiotemporal ocean-environment product that harmonizes bathymetry with data-driven currents and tides, with optional pollution fields for cleanup-related evaluation. On top of this environment representation, OneOcean defines a task ladder spanning navigation and station-keeping under currents, waypoint and route following, depth-profile tracking, area scanning, pipeline inspection, and multi-agent coordination, including formation transit, fish protection and patrol (herding), and surface pollution localization, containment, and cleanup. The suite supports multi-agent settings with 2–10 vehicles. Our simulator supports three to six degrees of freedom (DoF) and produces standardized metrics and reproducible run manifests, enabling consistent aggregation across tasks, difficulties, and scaling settings. Experiments across dataset variants and scenes systematically stress planning and control under data-grounded disturbances and reveal trade-offs among heuristic baselines, behavior cloning (BC), and a large language model (LLM)-based planner.

Project — <https://nebulis-lab.github.io/OneOcean/>

Code — <https://github.com/NEBULIS-Lab/OneOcean>

Demo — <https://nebulis-lab.github.io/OneOcean/demo/>

Introduction

Marine robots must operate under complex, highly variable ocean conditions. In practice, **currents and tides** can dominate vehicle motion, while field trials remain expensive and time-limited and at-sea sensing and communication are constrained (Zereik et al. 2018). Simulation is therefore essential, but existing platforms often trade off between (i) visually realistic underwater worlds, (ii) high-throughput, reproducible experimentation, and (iii) **ocean-condition realism grounded in real data**.

We present **OneOcean**, a data-grounded simulation suite and benchmark for marine robotics. OneOcean is built around a unified spatiotemporal ocean-environment dataset in Network Common Data Form (NetCDF). It harmonizes bathymetry from the General Bathymetric Chart of the Oceans (GEBCO) (GEBCO Compilation Group 2025) with

data-driven currents and tides from the Copernicus Marine Service Global Ocean Physics Analysis and Forecast (GOPAF) product (Copernicus Marine Service 2024; Copernicus Marine Environment Monitoring Service (CMEMS) 2025). On top of this environment representation, OneOcean defines a **10-task ladder** spanning single-agent navigation and tracking, spatial sensing and coverage, and multi-agent coordination and pollution-related tasks, with support for teams of 2–10 vehicles. The same environment product is also packaged into multiple dataset variants so the benchmark can support fast local iteration, scene-oriented construction, and broader public release without changing task semantics. Figure 1 summarizes this organization.

OneOcean provides two complementary execution backends that share task definitions and evaluation semantics. The *core benchmark backend* is deterministic and throughput-first, producing standardized metrics and reproducible run manifests for consistent aggregation. The *external underwater-scene backend* reuses packaged scenes from an existing simulator stack and injects OneOcean’s data-grounded currents, enabling qualitative evaluation in visually realistic underwater settings while preserving the benchmark protocol. This split separates controlled large-scale benchmarking from scene-rich portability studies while grounding both backends in the same ocean data product. In summary, our contributions are:

- A unified, reusable ocean-environment data product and tooling for exporting data-grounded disturbance signals to simulators, with variants that support fast iteration, scene construction, and broader public release.
- A canonical 10-task benchmark suite with difficulty levels, standardized metrics, auditable run manifests, and support for 3–6 DoF vehicle dynamics, with 6DoF used for the main quantitative evaluation.
- Experiments across the two backends that reveal task difficulty, scaling, and planner-compute trade-offs under data-grounded disturbances.

Related Work

Classical simulators built on Gazebo (Koenig and Howard 2004), including UUV Simulator (Manhães et al. 2016) and UWSim (Prats et al. 2012), provide dynamics and sensor stacks commonly used in robotics workflows, but they are not

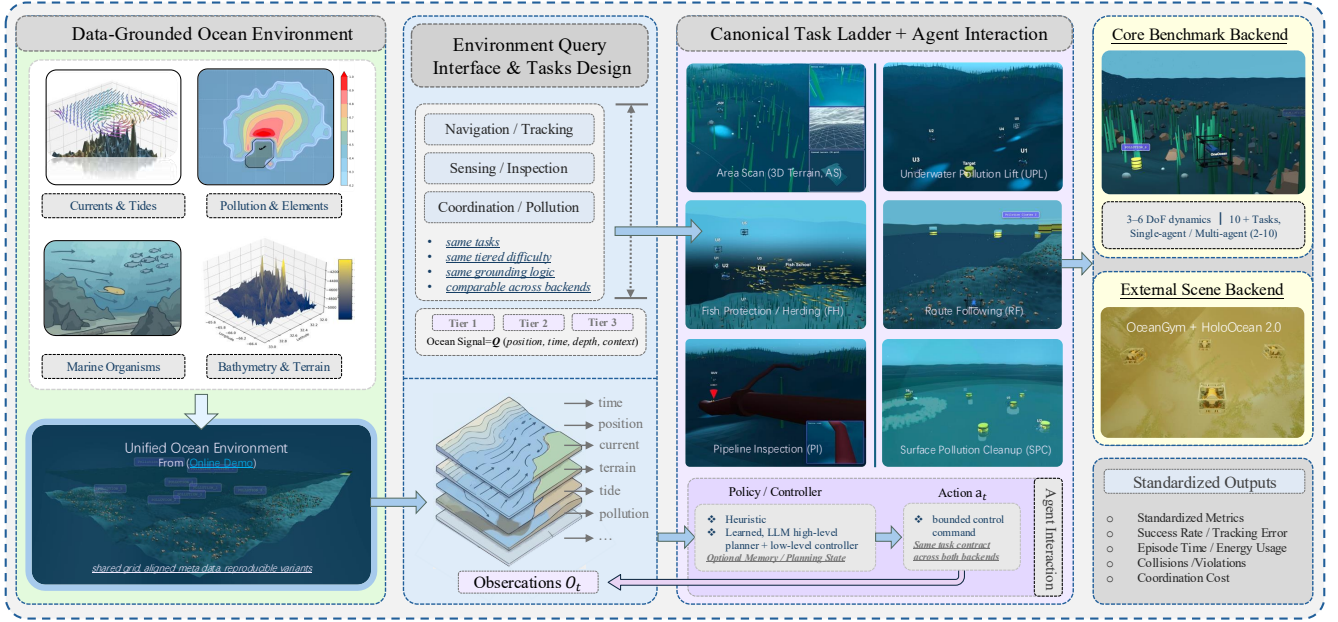


Figure 1: OneOcean overview. A unified ocean-environment product grounds a canonical multi-task benchmark and two complementary execution backends, enabling reproducible evaluation across navigation, sensing, coordination, and pollution-related marine-robotics tasks.

Symbols: ● Provided; ● Not provided. Tasks: G2G, SK, RF, DPT, AS, PI, FT, SPC, UPL, and FH.

Platform or Work	Venue	Real-data In Loop	Multi- agent	Cross- backend	3D Currents	Pollution or Spill	Tide Modeling	Canonical Task	Metrics Protocol	Headless Batch	G2G	SK	RF	DPT	AS	PI	FT	SPC	UPL	FH
MIMIR-UW	2023	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
HoloOcean 2.0	2025	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
OceanSim	2025	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
MarineGym	IROS 2025	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
OceanGym	2025	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
URLab	ICRA 2026	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
DREAM	ICRA 2026	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
Douro MARL	ICRA 2026	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
Map-TIDAL	ICRA 2026	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
CAMAR	AAAI 2026	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
RobotArena ∞	ICLR 2026	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
CaP-X	ICML 2026	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
OneOcean (ours)	AAAI 2027	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●

Table 1: System-level comparison and task coverage. Entries are conservatively coded from the cited papers and their public artifacts (Álvarez-Tuñón et al. 2023; Romrell et al. 2025; Song et al. 2025; Chu et al. 2025; Xue et al. 2025; Embley-Riches et al. 2026; Wu et al. 2026; Dal Fabbro et al. 2026; Ruediger and Banerjee 2026; Pshenitsyn, Panov, and Skrynnik 2026; Jangir et al. 2026; Fu et al. 2026).

designed for data-grounded ocean-condition benchmarking at scale. More recent systems target higher-fidelity rendering or higher-throughput experimentation, including MARUS (Lončar et al. 2022), OceanSim on Isaac Sim (Song et al. 2025; Makoviychuk et al. 2021), DAVE (Zhang et al. 2022), MarineGym (Chu et al. 2025), and URLab, which integrates Unreal Engine with MuJoCo (Embley-Riches et al. 2026). These platforms provide important simulator infrastructure, but they typically do not expose a reusable ocean-condition

product, a fixed task ladder, and a reporting protocol that remain consistent across multiple execution backends. Reproducing the same ocean-conditioned task across simulators therefore requires substantial custom engineering. Table 1 summarizes this distinction at both the system and task levels.

Ocean-condition realism is commonly provided by numerical models and operational forecasting systems. Representative community models include Delft3D (Lesser

et al. 2004), NEMO (Madec and the NEMO Team 2019), MITgcm (Strobach et al. 2018), ROMS (Shchepetkin and McWilliams 2005), and HYCOM (HYCOM Consortium for Data-Assimilative Ocean Modeling 2024). Operational products from the Copernicus Marine Service, including GOPAF, provide data-assimilative currents and hydrographic variables. GEBCO provides global gridded seafloor elevation, while NetCDF stores labeled multidimensional coordinates, variables, units, and provenance in a self-describing representation. These sources are public, globally scoped, and machine-queryable, allowing the same ocean condition to be extracted and audited across runs. Toolkits such as Parcels (Delandmeter and van Sebille 2019) and OpenDrift (Dagestad, Röhrs et al. 2018) support Lagrangian drift and dispersion analysis. OneOcean complements these efforts by treating ocean conditions as a reusable spatiotemporal product for robotics and pairing them with a protocol that runs across multiple simulator backends.

Recent task-specific marine systems provide complementary evidence. DREAM studies vision-language monitoring through simulation and tank trials (Wu et al. 2026); MapTIDAL allocates cooperative underwater-inspection tasks (Ruediger and Banerjee 2026); and Douro MARL maps a data-driven river plume with multiple vehicles (Dal Fabbro et al. 2026). Recent AI benchmarks broaden the evaluation perspective: CAMAR defines high-throughput continuous multi-agent routing and a three-tier protocol (Pshenitsyn, Panov, and Skrynnik 2026); RobotArena ∞ converts real demonstrations into scalable simulation tests for robot policies (Jangir et al. 2026); and CaP-X evaluates executable embodied agents across multiple manipulation suites (Fu et al. 2026). OneOcean addresses the remaining marine-specific gap by combining standardized environmental conditions, a canonical task suite, auditable run manifests, and shared contracts across two execution backends.

OneOcean Suite

OneOcean is organized around a single principle: treat *ocean conditions* as a reusable, auditable product, and evaluate robotics methods on a canonical task ladder under these conditions. This section summarizes the environment product, how it is consumed by the benchmark backends, and the task and evaluation definitions.

Unified spatiotemporal ocean-environment product

The shared NetCDF representation harmonizes GEBCO bathymetry with GOPAF currents and optional tides. It exposes named latitude, longitude, time, and depth coordinates, eastward and northward current components, elevation, and a mask for invalid terrain and NoData cells. The pipeline crops the requested region, retains the returned time and depth levels, aligns variables on one horizontal grid, and records provenance.

The core backend selects a fixed (t, d) slice for deterministic rollouts, whereas the external backend advances a dataset clock. Table 2 reports the realized products: the 0–200 m requests returned 26 levels through 186.13 m, while *tiny* contains one near-surface level at 0.494 m. Appendix

Variant	Region and period	$T \times D \times H \times W$	Resolution; size; use
<i>tiny</i>	42.1–42.7 N, 71.2–70.2 W; 2025	$357 \times 1 \times 62 \times 101$	$\approx 0.01^\circ$; 29 MB; tests and web
<i>scene</i>	32–33 N, 66.5–65.5 W; Dec. 2025	$23 \times 26 \times 240 \times 240$	0.00418° ; 690 MB; scenes
<i>public</i>	30–40 N, 72–62 W; 2025	$357 \times 26 \times 41 \times 41$	0.25° ; 175 MB; release

Table 2: Realized combined NetCDF variants. Sizes are the merged products themselves; source subsets and derived as-sets are distributed separately.

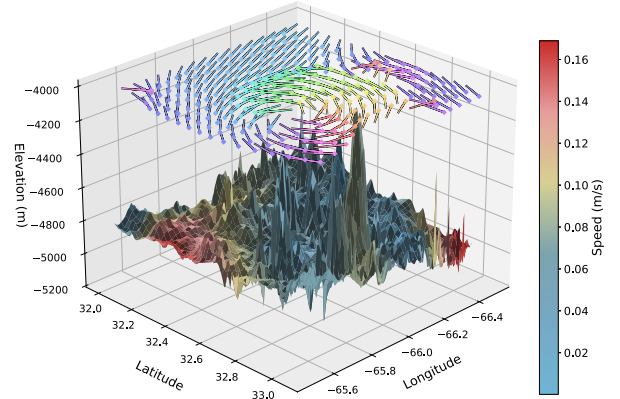


Figure 2: Base current field visualization over bathymetry. The horizontal current components are u_{data} and v_{data} ; surface color indicates speed magnitude (m s^{-1}). View configuration: pitch 25° , azimuth 45° , and depth index 0 m. The terrain surface is colored by speed and overlaid with arrows indicating current direction.

Sec. A specifies the full schema, mask semantics, coordinate conventions, validation results, and tide and depth caveats.

The mask denotes invalid terrain, crop-exterior, or NoData cells rather than a guaranteed coastline classification, and consumers must not interpret zero-filled masked elevation as seabed. Likewise, the single *tiny* layer is not evidence of resolved 3D flow. In the tide-enabled *scene* product, higher-frequency tide components are aligned to the base clock and attenuated with depth; this documented engineering profile supports controlled disturbance tests but not phase-accurate 3D tidal simulation.

Data-grounded disturbance signals

OneOcean exports either a fixed gridded slice for high-throughput rollouts or a temporal profile for scene-backed execution. Both apply current through a recorded relative-velocity convention in the core frame $(x, y, z) = (\text{east, depth-down, north})$, and every artifact records its extraction indices. Figure 2 visualizes one grounded current slice over bathymetry; Appendix Sec. A details the export and provenance contract.

Environment queries and probes. Shared queries expose local current, bathymetry, mask validity, and task signals.

Category	Task	Agents	Goal	Primary metric
Navigation	G2G	1	reach	final goal error
Navigation	SK	1	hold	station
				root-mean-square error
Navigation	RF	1	follow	cross-track error
Navigation	DPT	1	track	depth-tracking error
Sensing and Coverage	AS	1–10	scan	coverage fraction
Sensing and Inspection	PI	1–10	inspect	inspection score
Coordination	FT	2–10	transit	formation error
Pollution	SPC	2–10	clean	cleanup rate
Coordination	UPL	5	lift	lift success
Coordination	FH	8	herd	containment success

Table 3: Canonical task ladder used in OneOcean (10 tasks). Each task is evaluated at Tiers 1–3 and paired with a compact goal type and a primary task-specific metric.

Pollution tasks observe $o_i^c(t) = C(p_i(t), t)$ from either a current-advected Gaussian or a regeneratable Ocean Current Prediction Network (OCPNet) field; these are controlled signals, not in-situ labels. Appendix Sec. C gives the probe, sink, and discrete-source cleanup semantics.

Task ladder and difficulty levels

OneOcean defines ten tasks spanning navigation and tracking (G2G, SK, RF, DPT), sensing and inspection (AS, PI), and coordination and pollution response (FT, SPC, UPL, FH). Table 3 gives team sizes and primary metrics; Appendix Sec. B provides the complete objective, observation, constraint, and success contracts.

Tiers 1, 2, and 3 jointly increase current strength, precision, route and coverage extent, sensing difficulty, and coordination tightness while preserving each task’s interface. Every task declares its objective, observations, feasibility constraints, success condition, and primary metric. The release also includes a lightweight Robot Operating System 2 (ROS 2) bridge for state, probes, and velocity commands; its frame, topic, and service contract appears in Appendix Sec. C, while ROS performance remains outside this evaluation.

Navigation and tracking tasks test disturbance rejection and sustained trajectory accuracy; sensing and inspection tasks require spatial coverage or physical cue encounters; coordination and pollution tasks add team geometry, assignment, and dwell-time coupling. This grouping keeps the shared reporting contract compact without collapsing task-specific progress into success alone.

Two execution backends

OneOcean provides two complementary execution backends that share task and reporting contracts. The **core benchmark backend** is headless and throughput-first: it consumes a NumPy archive (NPZ) slice plus task and action configuration and returns observations, metrics, manifests, and recorder streams at scale. It supports 3–6 DoF engineering

dynamics (6DoF in the main experiments) (Manhães et al. 2016). The **external underwater-scene backend** prioritizes scene reuse: it consumes a temporal current export and the same task semantics inside packaged underwater scenes, returning quantitative route and task outputs with curated media. Thus, the second backend studies portability and scene realization; it is not claimed as a new rendering engine or as an architecture-matched throughput comparison.

Benchmark protocol and metrics. Actions specify desired relative velocity, clipped as $a_t = \text{clip}(\tilde{a}_t; v_{\max})$. With current gain g ,

$$c_t = g[u_{\text{data}}(x_t, z_t), 0, v_{\text{data}}(x_t, z_t)] + c_{\text{tide}}(t), \quad (1)$$

where u_{data} and v_{data} are sampled horizontal currents and c_{tide} is an optional synthetic robustness signal. With body velocity $\nu = [\nu_\ell^\top, \nu_a^\top]^\top$, rotation R , and diagonal M, D, K_ℓ, K_a, K_d , the 6DoF engineering update is

$$\begin{aligned} M\dot{\nu} &= \tau - D\nu, & \tau_\ell &= K_\ell(\nu_\ell^{\text{cmd}} - \nu_\ell), \\ \tau_a &= K_a(\nu_a^{\text{cmd}} - \nu_a) - K_d\nu_a, & \dot{p} &= R\nu_\ell + c_t. \end{aligned} \quad (2)$$

Commands are transformed to the body frame; yaw follows horizontal motion while roll and pitch are stabilized. Kinematic mode directly integrates $a_t + c_t$, and 3DoF suppresses roll and pitch. The diagonal model is not vehicle-identified hydrodynamics: it omits added mass, Coriolis and restoring terms, thruster allocation, and two-way fluid coupling. Hard terrain feasibility is

$$C = \{(x, y, z) \mid m_{\text{land}}(x, z) < \tau_{\text{land}}, y + \delta_{\text{clr}} \leq -h(x, z)\}, \quad (3)$$

where m_{land} is mask validity, h is bathymetric elevation, and δ_{clr} is required clearance. Masked, out-of-bounds, or clearance-violating proposals are rejected through the shared constraint path. We report SR, timeout-aware episode time T , $E = \sum_t \|a_t\|_2^2 \Delta t$, near-collision ratio, and task-specific metrics.

BC maps normalized goal, depth, probe, current, and task features ϕ_t to $\hat{a}_t = f_\theta(\phi_t)$ and minimizes the heuristic-action loss

$$\mathcal{L}_{\text{BC}} = N_{\text{BC}}^{-1} \sum_t \|a_t^{\text{heur}} - \hat{a}_t\|_2^2. \quad (4)$$

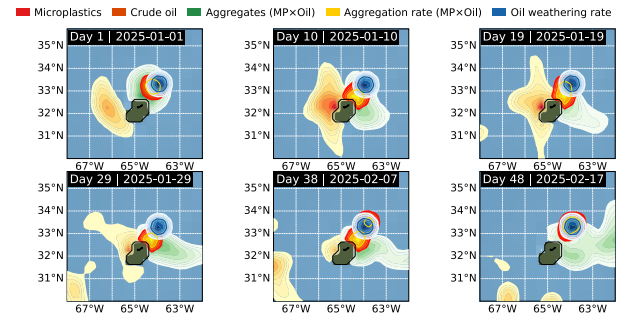


Figure 3: Curated OCPNet multi-species diffusion and reaction field used as qualitative evidence. We treat these fields as regeneratable signals that can be coupled to localization, containment, and cleanup tasks through probes and task-specific metrics.

Method	Macro (Tier 2)			Macro (Tier 3)			Tier 3 key-task metrics					
	Suite SR(%)	E	Coll. (%)	Suite SR(%)	E	Coll. (%)	G2G SR(%)	RF SR(%)	SK RMS	Scan cov.	Pipe score	Cleanup rate
Heuristic	84.6	1842.3	5.68	29.0	3352.4	8.19	74.0	54.0	1.07	0.89	0.78	0.56
BC	58.6	1271.2	9.72	6.0	1635.1	6.41	34.0	0.0	26.52	0.89	0.67	0.06

Table 4: Headline core-benchmark results (6DoF): macro averages at Tiers 2 and 3 plus Tier 3 key-task metrics, using 50 episodes per task and tier. Success rate (SR) is reported in percent; Macro Suite SR is an unweighted mean across the 10 canonical tasks. Coll is the mean collision-step ratio (%) with a 2.0 m radius. Best values per column are highlighted.

Method	Avg. SR(%)	Cleanup SR(%)	Cleanup rate	Scan SR(%)	Scan cov.	Pipe SR(%)	Pipe score	$T(s)$	E
Heuristic	65.8	60.0	0.87	77.5	0.80	60.0	0.69	776.8	9132.4
BC	45.0	10.0	0.32	67.5	0.79	57.5	0.61	921.6	6908.2
LLM-ChatGLM3-6B	55.0	60.0	0.87	77.5	0.82	27.5	0.54	860.3	9671.5
LLM-Qwen2.5-7B	47.5	57.5	0.81	57.5	0.78	27.5	0.57	1077.2	11703.8
LLM-Qwen2-7B	45.0	57.5	0.86	67.5	0.81	10.0	0.51	991.4	10422.1
LLM-Mistral-7B	42.5	55.0	0.82	55.0	0.78	17.5	0.52	1044.2	11288.9
LLM-Llama3-8B	44.2	67.5	0.84	47.5	0.77	17.5	0.59	1099.1	11614.2
LLM-Llama2-7B	30.8	55.0	0.87	27.5	0.75	10.0	0.46	1156.4	11654.3

Table 5: Core-benchmark planning-suite performance at Tier 2 with doubled currents: heuristic vs BC vs local LLM planners on three planning-sensitive tasks, using 40 episodes per task (120 per method). Best values per column are highlighted.

Every K steps, an LLM instead proposes a discrete assignment $g_k = \pi_{\text{LLM}}(s_k)$; schema-invalid output invokes a deterministic allocator, and a shared low-level controller executes $a_t = \pi_{\text{low}}(x_t, g_k)$. Appendix Sec. C gives dynamics defaults, pollution semantics, and ROS mapping; Appendix Sec. D defines all metrics and manifests.

Optional pollution fields (OCPNet). Continuous containment applies a local concentration sink, whereas canonical SPC completes discrete sources after an assignment-and-dwell condition; its cleanup rate is therefore source completion, not removed fluid mass. Figure 3 shows the regeneratable multi-species signal.

Experiments

We evaluate OneOcean along four axes: method separation on the unified 10-task suite under 6DoF settings; difficulty and control-versus-planning gaps across the tiered task ladder; trade-offs in success, efficiency, and compute among heuristic control, BC, and local LLM planners; and transfer of the same task semantics to the external underwater-scene backend grounded by the same ocean data product.

Experimental setup

The main core results use 6DoF with hard mask and clearance checks and a fixed NetCDF-derived current slice; the external backend advances a temporal export in four packaged scenes. We report SR, timeout-aware T , action-energy proxy E , near-collision ratio, and the task metrics named in the tables. Headline budgets are stated with their result tables, while separate diagnostics audit planner decisions, dynamics

sensitivity, and pollution observations. All runs store their integer seed, task and dynamics configuration, dataset indices, and software and Git state. Local LLM inference assigns one 48 GB accelerator to each process. Appendix Sec. D specifies the supplementary diagnostic budgets and complete manifest contract.

Here T uses success time on successful episodes and time-out on failures; E is computed only from clipped vehicle commands and is not inference energy. Coll(%) counts steps in which the closest agent pair is within the configured near-collision radius. Coverage, inspection, source completion, station error, cross-track error (CTE), and reached waypoints expose progress that binary SR would hide. Tiers 1–3 jointly change disturbance, tolerance, spatial extent, and coordination settings rather than varying one scalar.

Core benchmark backend

Figure 4 shows four representative tasks from the canonical suite. The quantitative evidence in Tables 4 and 6 establishes two main properties of the benchmark. First, Tier 2 is challenging but not saturated for all methods, making it useful for regression testing and planner comparison. Second, Tier 3 exposes large method gaps on both single-agent and multi-agent tasks, especially in station keeping, route following, surface cleanup, and leak inspection.

Figure 5 makes the ladder effect explicit on a representative task subset: the left panel shows the tier-wise success profile, while the right panel isolates where the largest drops occur as the benchmark moves from Tier 1 to Tier 3.

Two observations stand out. Heuristic control remains the strongest general-purpose baseline in the core benchmark backend, especially once Tier 3 activates stricter tolerances

Task	N	Heur T2	BC T2	Heur T3	BC T3
<i>Navigation and Tracking</i>					
G2G	1	94.0	44.0	74.0	34.0
SK	1	100.0	0.0	14.0	0.0
RF	1	84.0	24.0	54.0	0.0
DPT	1	84.0	64.0	50.0	0.0
<i>Coordination</i>					
FT	10	100.0	40.0	20.0	0.0
FH	8	94.0	100.0	4.0	4.0
<i>Sensing and Pollution</i>					
PI	1	100.0	90.0	30.0	14.0
AS	1	90.0	84.0	14.0	4.0
SPC	10	84.0	50.0	10.0	0.0
UPL	5	16.0	90.0	20.0	4.0

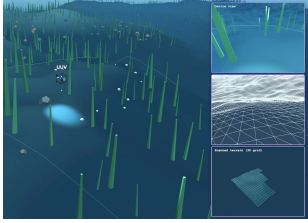
Table 6: Per-task success rates at Tier 2 and Tier 3 for the canonical 10-task suite, using 50 episodes per task and tier. Highlighted rows are revisited in the planning and external-backend transfer studies.



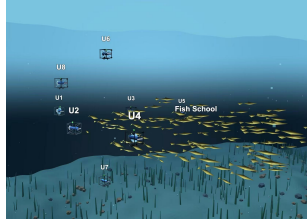
(a) SPC: Surface Pollution Cleanup (multi-agent)



(b) UPL: Underwater Pollution Lift to Surface (5 UUVs)



(c) AS: Area Scan over terrain



(d) FH: Fish Protection and Herding (8 UUVs)

Figure 4: Representative core-benchmark tasks used in this work’s experiments. These snapshots connect the quantitative tables to the task geometry and multi-agent interactions used in the benchmark.

and stronger drift. BC remains competitive on *Underwater-Lift* and *Fish-Herding* at Tier 2. However, it degrades sharply on tasks that require persistent disturbance rejection or coordinated geometry maintenance. Figure 6 further shows that the surface-cleanup task captures a meaningful team-size trade-off: larger teams improve success and source-completion rate, but collision risk increases and the energy budget does not scale monotonically.

Method	Avg. SR(%)	Calls	Lat. (s)	ms per call	Prompt tok.	Output tok.	Total tok.
LLM-Qwen2.5-7B	65.0	6.48	14.21	2222.1	3932	4084	8016
LLM-Qwen2-7B	53.3	4.71	11.24	2440.3	2877	2996	5873
LLM-Mistral-7B	53.3	6.05	31.38	4942.5	3762	4128	7890
LLM-ChatGLM3-6B	53.3	5.38	7.91	1393.6	3414	411	3825
Heuristic	55.0	0.00	0.00	0.0	0	0	0
LLM-Llama3-8B	45.0	6.71	13.94	2117.5	3072	3218	6292
LLM-Llama2-7B	33.3	6.59	49.42	7401.8	4219	4851	9071
BC	21.7	0.00	0.00	0.0	0	0	0

Table 7: Core-benchmark planning-suite cost metrics at Tier 2 with doubled currents: uncached local-inference calls, latency, and token usage, reported alongside average success over 20 episodes per task.

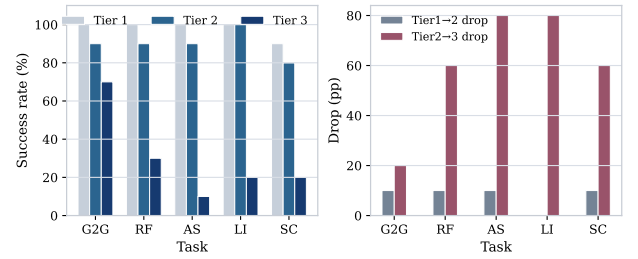


Figure 5: Difficulty-ladder summary on a representative five-task subset. Left: success rate at Tiers 1–3. Right: the corresponding success-rate drops from Tier 1→2 and Tier 2→3, highlighting that the hardest setting remains strongly discriminative for route following, area scan, leak inspection, and surface cleanup.

Planning-sensitive tasks. We next focus on three planning-sensitive tasks: surface cleanup, area scan, and leak inspection. We compare deterministic heuristic control, BC, and several local LLM planners. The LLM never replaces low-level control; it proposes high-level assignments or way-point choices at a fixed stride, while the low-level controller and safety checks remain deterministic.

Tables 5 and 7 show a useful separation. The heuristic remains the most reliable overall planner, but several local LLMs are competitive on specific tasks: ChatGLM3-6B matches the heuristic on cleanup rate and scan coverage with lower latency than the other LLMs, while Qwen2.5-7B gives the best average success among the LLM family in this sweep.

An instrumented three-seed audit explains why valid language output does not imply task success: fallback can invoke a competent allocator, while valid plans can still repeat covered regions or miss leaks. Thus E measures executed trajectories, whereas latency and token counts measure inference cost. The LLM rows are high-level planning diagnostics rather than architecture-matched low-level controller rankings. Appendix Sec. E reports task-level and model-level validity, fallback, completion, token, and latency values.

Scene	Tier 2			Tier 3		
	SR (%)	WP	CTE (m)	SR (%)	WP	CTE (m)
Dam	0.0	1.0	3.06	0.0	0.0	0.87
OpenWater	20.0	3.1	2.50	0.0	0.0	1.91
PierHarbor	56.7	3.5	2.22	0.0	2.0	1.41
SimpleUnderw.	100.0	4.0	3.04	96.7	3.0	2.16
Mean over scenes	44.2	2.92	2.70	24.2	1.25	1.59

Table 8: Route-following across four underwater scenes after the success-criterion fix. Each scene uses 30 episodes per difficulty. “Waypoints” is the mean number of reached waypoints and CTE is cross-track error.

External underwater-scene backend

Figure 7 and Table 8 show that the shared task contract remains executable but route following degrades differently across scenes. A separate diagnostic found lower localization error with OCPNet-3D under both tested current settings without reducing aggregate success. Appendix Sec. F provides the values, episode budget, and interpretation limits.

The external underwater-scene results should be read as a portability study rather than as a replacement for core-benchmark throughput evaluation. This backend reuses realistic scenes and produces paper and web media, but it also introduces stricter scene geometry and a different disturbance hook. Both backends retain the same task semantics, data provenance, and reporting contract.

Discussion and Limitations

OneOcean’s data-grounded disturbances provide repeatable stress tests across task families. Tier 2 supports regression testing, whereas Tier 3 exposes failures in tracking and coordination. No method dominates every criterion: heuristic control is strongest overall, BC remains competitive on selected structured tasks, and local LLM planners trade improved planning behavior for additional latency and token cost.

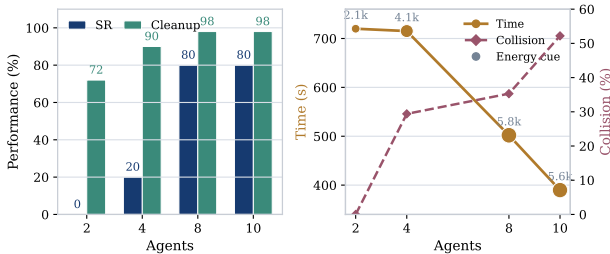


Figure 6: Surface-cleanup scaling at Tier 2. Left: success rate and cleanup rate as the number of vehicles increases. Right: completion time and near-collision ratio, with per-point annotations showing the energy proxy in thousands. The figure highlights that scaling improves cleanup outcomes, but also shifts the coordination cost profile.



(a) Surface pollution containment and cleanup in a packaged ocean scene. (b) Multi-agent scene-backed deployment.

Figure 7: Qualitative evidence from the external underwater-scene backend, which reuses packaged underwater worlds while injecting currents exported from the same combined dataset used by the core benchmark backend.

Success alone can obscure substantial partial progress and differences in energy, collision exposure, or completion time. We therefore pair success with task-specific progress. Planner validity is reported separately from closed-loop performance because a syntactically valid assignment may be dynamically poor, while deterministic fallback can conceal a failed planner call. Backend portability is assessed through shared task contracts, data provenance, output semantics, and common constraint checks rather than direct visual-fidelity or throughput rankings between unlike simulators. A controller, vehicle, or backend is compatible only when it consumes the environment query contract, declares termination conditions and metrics, records its configuration and dataset indices, and preserves the shared constraint path.

The validator and ROS 2 smoke checks establish interface compliance rather than physical fidelity or task performance. OneOcean remains a controlled engineering abstraction rather than a full hydrodynamic simulator. Its diagonal 6DoF update omits added mass, Coriolis and restoring forces, thruster allocation, and two-way fluid coupling. Dataset currents use a recorded relative-transport convention; the external backend applies them through a one-way disturbance hook. The `tiny` product has one near-surface layer, whereas the larger products retain 26 measured horizontal-current levels. OCPNet nevertheless repeats horizontal current over model depth and assumes zero vertical current. Appendix Sec. D details the remaining pollution, tide, ROS 2, and extension boundaries.

Conclusion

We introduced OneOcean, a data-grounded simulation suite and benchmark for marine robotics built around a unified spatiotemporal ocean-environment product, a canonical 10-task ladder, and standardized metrics and manifests. Across a throughput-first headless backend and an external underwater-scene backend, OneOcean supports reproducible evaluation under data-grounded disturbances and exposes trade-offs between heuristic control, BC, and local LLM planners. The experiments establish its utility for controlled quantitative evaluation, robustness analysis, planner-cost analysis, and scene-backed portability studies. Future work will improve tide coupling and three-dimensional cur-

rent structure, broaden task families and robot morphologies, and strengthen public release workflows for derived datasets and benchmark artifacts.

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